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ABSTRACT

Package-level integration using multi-chip-modules (MCMs) is a promising approach for building large-scale systems. Compared to a large monolithic die, an MCM combines many smaller chiplets into a larger system, substantially reducing fabrication and design costs. Current MCMs typically only contain a handful of coarse-grained large chiplets due to the high area, performance, and energy overheads associated with inter-chiplet communication. This work investigates and quantifies the costs and benefits of using MCMs with fine-grained chiplets for deep learning inference, an application area with large compute and on-chip storage requirements. To evaluate the approach, we architected, implemented, fabricated, and tested Simba, a 36-chiplet prototype MCM system for deep-learning inference. Each chiplet achieves 4 TOPS peak performance, and the 36-chiplet MCM package achieves up to 128 TOPS and up to 6.1 TOPS/W. The MCM is configurable to support a flexible mapping of DNN layers to the distributed compute and storage units. To mitigate inter-chiplet communication overheads, we introduce three tiling optimizations that improve data locality. These optimizations achieve up to 16% speedup compared to the baseline layer mapping. Our evaluation shows that Simba can process 1988 images/s running ResNet-50 with batch size of one, delivering inference latency of 0.50 ms.

CCS CONCEPTS

• **Computer systems organization** → *Interconnection architectures; Multicore architectures; Neural networks; Data flow architectures; Special purpose systems.*

KEYWORDS

Multi-chip module, neural networks, accelerator architecture

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1 INTRODUCTION

Deep learning (DL) [44] has become critical for addressing complex real-world problems. In particular, deep neural networks (DNNs) have demonstrated their effectiveness across a wide-range of applications, including image recognition [33, 41, 60, 64, 65], object detection [27, 54], language translation [63, 70], audio synthesis [69], and autonomous driving [10]. State-of-the-art DNNs [6, 12, 27, 33, 41, 46, 54, 60, 64, 65] require billions of operations and hundreds of megabytes to store activations and weights. Given the trend towards even larger and deeper networks, the ensuing compute and storage requirements motivate large-scale compute capability in DL hardware, which is currently addressed by a combination of large monolithic chips and homogeneous multi-chip board designs [14, 17, 24, 29, 39, 71]. Previously proposed multi-chip DL accelerators have focused on improving total compute throughput and on-chip storage size but have not addressed the scalability challenges associated with building a large-scale system with multiple discrete components.

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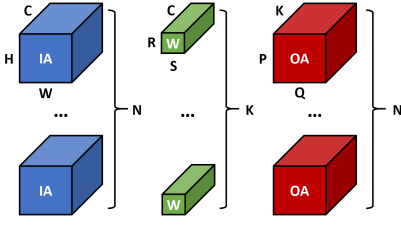


Figure 1: Input activations (IA), weights (W), and output activations (OA) in convolutional layers.

Recently, the need for high compute throughput in an era of slowing transistor scaling has motivated advances in multi-chip-module (MCM) integration to build large-scale CPUs [7, 37, 40, 62] and GPUs [3, 74]. MCM packaging approaches can also reduce cost by employing smaller chiplets connected together post-fabrication, as yield losses cause fabrication cost to grow super-linearly with die size. Packaging technologies including organic substrates [36] and silicon interposers [28, 55] can be used to assemble a large-scale MCM system. In addition, recent advances in package-level signaling offer the necessary high-speed, high-bandwidth signaling needed for chiplet-based system [72]. As a result, chiplet-based systems using MCMs can provide improved performance more efficiently than board-level integration but with lower cost than monolithic chips. While MCMs have been used for general compute systems, applying MCMs to high-performance DNN inference algorithms has not been previously examined. Specific challenges stem from the natural non-uniformity between on-chip and on-package bandwidth and latency. While multi-chip systems also exhibit similar forms of non-uniformity, this paper focuses on the specific characteristics of MCM-based systems as they provide a natural progression from monolithic single-chip inference accelerators as semiconductor scaling slows.

This paper presents Simba, a scalable deep-learning inference accelerator employing multi-chip-module-based integration. Each of the Simba chiplets can be used as a standalone, edge-scale inference accelerator, while multiple Simba chiplets can be packaged together to deliver data-center-scale compute throughput. We specifically examine the implications of the non-uniform latency and bandwidth for on-chip and on-package communication that lead to significant latency variability across chiplets. Such latency variability results in a long “tail latency” during the execution of individual inference layers. As a result, the overall performance for each layer is limited by the slowest chiplet in the system, limiting scalability. To address these challenges, we propose three tail-latency-aware, non-uniform tiling optimizations targeted at improving locality and minimizing inter-chiplet communication: (1) non-uniform work partitioning to balance compute latency with communication latency; (2) communication-aware data placement to minimize inter-chiplet traffic; and (3) cross-layer pipelining to improve resource utilization.

To explore the challenges and evaluate the benefits of MCM-based inference accelerator architectures, we designed, implemented, and fabricated a prototype of Simba, consisting of 36 chiplets connected in a mesh network in an MCM [76]. The 6 mm² chiplets are

```

1 for n = [0 : N) :
2   for p = [0 : P) :
3     for q = [0 : Q) :
4       for k = [0 : K) :
5         for r = [0 : R) :
6           for s = [0 : S) :
7             for c = [0 : C) :
8               OA[n,p,q,k] += IA[n,h,w,c] * W[k,r,s,c]
```

Listing 1: DNN loop nest.

fabricated in a 16 nm FinFET process technology and contain both multiply-accumulate (MAC) units and an SRAM-based memory system. Each chip has a peak performance of 4 tera-ops per second (TOPS) using 8-bit weights and activations and 24-bit accumulation. The 36-chiplet MCM achieves up to 128 TOPS with an energy efficiency range of 0.2–6.1 TOPS/W depending on operating voltage. Simba supports flexible mapping and resource allocation for efficient DNN inference execution across a wide range of workloads. We thoroughly characterize the Simba system and motivate the importance of task and data placement in the Simba MCM system, which can lead to as much as a 2.5× performance difference for individual ResNet-50 layers. Motivated by our observations, we propose non-uniform work and data placement, together with cross-layer pipelining, to improve system utilization in the presence of small batch sizes and communication latency.

2 BACKGROUND AND MOTIVATION

Many applications from edge devices to data centers demand fast and efficient inference, often with low latency or real-time throughput requirements. Today’s DNN inference applications typically run on highly programmable but inefficient CPU-based systems, programmable GPUs with ISA extensions for accelerating tensor operations, or fixed-function DNN inference accelerators. Recent work has shown that fixed-function accelerators can provide orders of magnitude better energy efficiency and performance than CPUs and better area and energy efficiency than GPUs [1, 13–15, 20, 25, 30, 51, 59, 75].

Small-scale inference applications can run on moderately-sized chips, while those demanding higher performance may require large monolithic chips or board-level multi-chip solutions to scale. Partitioning an application across multiple chips at the board level is not easy because of the enormous difference in bandwidth, latency, and energy between on-chip communication and inter-chip communication. Other packaging approaches such as multi-chip modules can provide inter-chip interconnect that is closer in nature to on-chip interconnect, offering a more straightforward path to scaling.

2.1 DNN Basics

DNNs are constructed using a series of layers, including convolutional layers, pooling layers, activation layers, and fully-connected layers. A convolutional layer is algorithmically formulated as a seven-dimensional nested loop over an input activation (IA) tensor, a weight (W) tensor, and an output activation (OA) tensor, as shown in Figure 1. Listing 1 shows the convolution computation embedded in a seven-dimensional loop nest. The same formulation

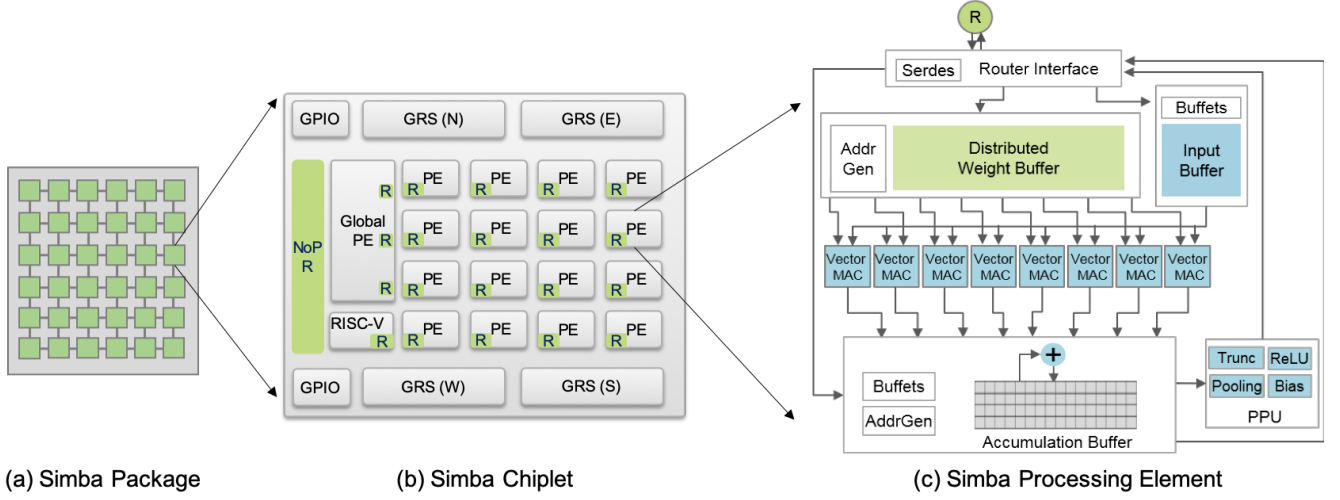


Figure 2: Simba architecture from package to processing element (PE).

also applies to fully-connected layers that are widely used in multi-layer perceptrons (MLPs) and recurrent neural networks (RNNs). An activation layer applies a non-linear function such as *ReLU* or *sigmoid*, while pooling layers down-sample the input activations after convolutional layers. Activation and pooling layers are typically merged with convolutional layers during execution to reduce data movement.

2.2 Multi-Chip-Module Packaging

Package-level MCM integration is a promising alternative for assembling large-scale systems out of small building blocks known as *chiplets*. Such systems consist of multiple chiplets connected together via on-package links using a silicon interposer or an organic substrate and employing efficient intra-package signaling circuits [7, 21, 40, 72]. Compared to a large monolithic die, MCMs can reduce (1) design costs, since logic design, verification, and physical design are all easier on a small chip than a large chip; and (2) fabrication costs, as the much lower manufacturing yield of large chips make them far more expensive than small chips. In addition, different scales of systems can be created merely by adjusting the number of chiplets placed in a package, without requiring a different chip tapeout for each market segment. MCMs have been recently applied to a general-purpose CPU design [7] as an alternative to building multi-core CPUs on reticle-limited large die. They have also been an active research area for scaling of multi-CPU [37, 40, 45, 62] and multi-GPU systems [3, 18, 74]. However, package-level wires do not provide the same communication density or energy/bit as on-chip wires. Consequently, MCM architects and software developers must still consider the non-uniform bandwidth, latency, and energy present in these systems to achieve efficient application performance.

2.3 Non-Uniformity in MCM-based Design

An MCM-based system has a heterogeneous interconnect architecture, as the available intra-chiplet bandwidth is expected to be

significantly higher than available inter-chiplet bandwidth. In addition, sending data to remote chiplets incurs additional latency. This latency may include on-chip wire delays to move data to the edge of the chiplet, synchronizer delays for crossing clock domains, serialization and deserialization latency in high-speed communication links, and the on-package wire delays of inter-chiplet links. As a result, communication latency between two elements in a MCM heavily depends on their spatial locality on the package.

Mapping DNN layers to a tile-based architecture is a well-studied research problem [15, 25, 50]. State-of-the-art DNN tiling typically assumes a flat architecture with uniform latency and bandwidth across processing elements and focuses on data reuse for reducing global bandwidth demands. This assumption is acceptable for small-scale systems, as the communication latency variability is small, and the computation is often tolerant of communication latencies. However, as DNN inference performance is scaled up to larger systems, the execution time decreases and latency-related effects become more important. Furthermore, in large-scale systems with heterogeneous interconnect architectures such as MCMs, assumptions of uniform latency and bandwidth in selecting DNN tiling can degrade performance and energy efficiency. Simba is the first work that quantitatively highlights the challenge of mapping DNN layers to non-uniform, MCM-based DNN accelerators and proposes communication-aware tiling strategies to address the challenge.

3 SIMBA ARCHITECTURE AND SYSTEM

To understand the challenges and opportunities of using MCMs for building large-scale, deep-learning systems, we designed, implemented, fabricated, and characterized Simba, the first chiplet-based deep-learning system. This section first presents an overview of the Simba architecture and its default uniform tiling strategy. We then describe Simba's silicon prototype and present a detailed characterization of the Simba system in Section 4.

Table 1: Simba system communication capability.

Packet Source	Unicast Destination	Multicast Destination
PE	Local PEs, Global PE, Controller	-
	Remote PEs, Global PE, Controller	-
Global PE	Local PEs, Controller	Local PEs
	Remote PEs, Controller	Remote PEs
Controller	Local PEs, Global PE	
	Remote PEs, Global PE, Controller	

3.1 Simba Architecture

Tile-based architectures have frequently been proposed for deep-learning accelerator designs [1, 2, 13, 15, 22, 25, 30, 42, 51, 53, 56, 58, 61, 75]. Our design target is an accelerator scalable to data center inference, where state-of-the-art data center accelerators deliver around 100 tera-operations-per-second (TOPS). For example, the first generation of the Tensor Processing Unit (TPU) delivers 92 TOPS [39] and is designed for inference applications. One simple approach to achieve this design goal is to increase the number of tiles in a monolithic single chip. However, building a flat network with hundreds of tiles would lead to high tile-to-tile communication latency, as examined in both multi-core CPU [19] and accelerator [26] research.

Simba adopts a hierarchical interconnect to efficiently connect different processing elements (PEs). This hierarchical interconnect consists of a network-on-chip (NoC) that connects PEs on the same chiplet and a network-on-package (NoP) that connects chiplets together on the same package. Figure 2 illustrates the three-level hierarchy of the Simba architecture: package, chiplet, and PE. Figure 2(a) shows a Simba package consisting of a 6×6 array of Simba chiplets connected via a mesh interconnect. Each Simba chiplet, as shown in Figure 2(b), contains an array of PEs, a global PE, a NoP router, and a controller, all connected by a chiplet-level interconnect. To enable the design of a large-scale system, all communication between the PEs, Global PEs, and controller is designed to be latency-insensitive [11] and is sent across the interconnection network through the NoC/NoP routers.

Simba PE: Figure 2(c) shows the microarchitecture of the Simba PE, which includes a distributed weight buffer, an input buffer, parallel vector MAC units, an accumulation buffer, and a post-processing unit. Each Simba PE is similar to a scaled-down version of NVDLA, a state-of-the-art DL accelerator product [59]. The heart of the Simba PE is an array of parallel vector multiply-and-add (MAC) units that are optimized for efficiency and flexibility. The Simba PE uses a weight-stationary dataflow: weights remain in the vector MAC registers and are reused across iterations, while new inputs are read every cycle. Each vector MAC performs an 8:1 dot-product along the input channel dimension C to exploit an efficient spatial reduction [42]. To provide flexible tiling options, the Simba PE also supports cross-PE reduction with configurable producers and consumers. If the current PE is the last PE on the reduction chain, it first sends partial sums to its local post-processing unit that performs ReLU, truncation and scaling, pooling, and bias addition. The final

output activation is sent to the target Global PE for computation of the next layer.

Simba Global PE: The Global PE serves as second-level storage for input/output activation data to be processed by the PEs. To support flexible partitioning of the computation, the Global PE can either unicast data to one PE or multicast to multiple PEs, even across chiplet boundaries. The Global PE has a multicast manager that oversees these producer-consumer relationships. The Global PE also serves as a platform for near-memory computation. Many DNNs feature some computation that has low data reuse, such as element-wise multiply/add in ResNet [33] or depth-wise convolution in MobileNet [34]. The Global PE can perform such computations locally to reduce communication overhead for these types of operations.

3.2 Simba Silicon Prototype

Simba Controller: Each Simba chiplet contains a RISC-V processor core [4] that is responsible for configuring and managing the chiplet’s PEs and Global PE states via memory-mapped registers using an AXI-based communication protocol. After all states are configured, the RISC-V triggers execution in the active PEs and Global PEs and waits for these blocks to send *done* notifications via interrupts. Synchronization of chiplet control processors across the package is implemented via memory-mapped interrupts.

Simba Interconnect: To efficiently execute different neural networks with diverse layer dimensions, Simba supports flexible communication patterns across the NoC and NoP. Table 1 lists Simba communication capability across all components. Both NoC and NoP use a mesh topology with a hybrid wormhole/cut-through flow control. Specifically, unicast packets use wormhole flow control for large packet size, while multicast packets are cut-through to avoid wormhole deadlocks. Each Simba PE can unicast to any local or remote PE for cross-PE partial-sum reduction, to any local or remote Global PE to transmit output activation values, and to any local or remote chiplet controller to signal execution completion. A PE does not need to send multicast packets as its computation requires only point-to-point communication. In addition to unicast communication, a Global PE can also send multicast packets to local and remote PEs for flexible data tiling.

We implemented, fabricated, and tested a silicon prototype of the Simba system, shown in Figure 3, with the microarchitecture

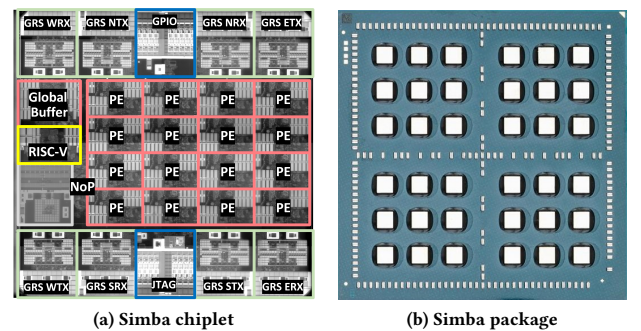
**Figure 3: Simba silicon prototype.**

Table 2: Simba microarchitecture parameters.

Package	Number of Chiplets	36
	Size	47.5 mm × 47.5 mm
	Core Voltage	0.52–1.1 V
	PE Clock Frequency	0.48–1.8 GHz
	Chiplet-to-Chiplet Interconnect	Ground-Referenced Signaling
	NoP Interconnect Bandwidth	100 GB/s/Chiplet
	NoP Interconnect Latency	20 ns/Hop
	NoP Interconnect Energy	0.82–1.75 pJ/bit
Chiplet	Number of PEs	16
	Area	2.5 mm × 2.4 mm
	Technology	16 nm FinFET
	Voltage	0.42–1.2 V
	PE Clock Frequency	0.16–2.0 GHz
	Global PE Buffer Size	64 KiB
	Routers Per Global PE	3
	NoC Interconnect Bandwidth	68 GB/s/PE
	NoC Interconnect Latency	10 ns/Hop
	Microcontroller	RISC-V
PE	Weight Buffer Size	32 KiB
	Input Buffer Size	8 KiB
	Accumulation Buffer Size	3 KiB
	Vector MAC Width	8
	Number of Vector MACs	8
	Dataflow	Weight Stationary
	Input/Weight Precision	8 bits
	Partial-Sum Precision	24 bits

parameters in Table 2. We chose parameters so that a Simba chiplet has area and power similar to an efficient edge system, such as Dian-Nao [13] or Eyeriss [15], while a full Simba package is comparable to a data-center-scale system such as TPU [39]. Table 3 shows the synthesis area breakdown of key components in the Simba chiplet architecture.

Shown in Figure 3a, the 2.5 mm × 2.4 mm Simba chiplets were implemented in TSMC 16 nm FinFET process technology [76]. Each Simba package (Figure 3b) contains an array of 6 × 6 chiplets connected on an organic package substrate using ground-referenced signaling (GRS) technology for intra-package communication [72]. The top and bottom rows of each chiplet include eight chiplet-to-chiplet GRS transceiver macros. Four macros are configured as receivers and four as transmitters. Each transceiver macro has four data lanes and a clock lane with configurable speed from 11 Gbps/pin to 25 Gbps/pin, consuming 0.82–1.75 pJ/bit, with a total peak chiplet bandwidth of 100 GB/s. We chose GRS as our communication mechanism because it delivers 3.5× higher bandwidth per unit area and lower energy per bit compared to other MCM interconnects [7].

The prototype chiplets were implemented using a globally asynchronous, locally synchronous (GALS) clocking methodology [23],

Table 3: Area breakdown of the Simba system.

Partition	Component	Area (μm ²)
PE	Vector MACs	12K
	Weight Buffer	41K
	Input Buffer	11K
	Accumulation Buffer	24K
	NoC Router	19K
Global PE	Distributed Buffer	125K
	NoC Routers	27K
RISC-V	Processor	109K
NoP	NoP Router	42K

```

1 //Package level
2 for p3 = [0 : P3) :
3   for q3 = [0 : Q3) :
4     parallel_for k3 = [0 : K3) :
5       parallel_for c3 = [0 : C3) :
6         // Chiplet level
7         for p2 = [0 : P2) :
8           for q2 = [0 : Q2) :
9             parallel_for k2 = [0 : K2) :
10              parallel_for c2 = [0 : C2) :
11                // PE level
12                for r = [0 : R) :
13                  for s = [0 : S) :
14                    for k1 = [0 : K1) :
15                      for c1 = [0 : C1) :
16                        for p1 = [0 : P1) :
17                          for q1 = [0 : Q1) :
18                            // Vector-MAC level
19                            parallel_for k0 = [0 : K0) :
20                              parallel_for c0 = [0 : C0) :
21                                p = (p3 * P2 + p2) * P1 + p1;
22                                q = (q3 * Q2 + q2) * Q1 + q1;
23                                k = ((k3 * K2 + k2) * K1 + k1) * K0 + k0;
24                                c = ((c3 * C2 + c2) * C1 + c1) * C0 + c0;
25                                OA[p,q,k] += IA[p-1+r,q-1+s,c] * W[r,s,c,k];

```

Listing 2: Simba baseline dataflow.

allowing independent clock rates for individual PEs, Global PEs, RISC-V processors, and NoP routers. Running in a single-chiplet configuration, Simba prototypes have been measured to operate correctly in the lab at a minimum voltage of 0.42 V with a 161 MHz PE frequency, achieving 0.11 pJ/Op (9.1 TOPS/W) core power efficiency on a peak-utilization convolution micro-benchmark. At 1.2 V, each chiplet operates with a 2 GHz PE frequency for a peak throughput of 4 TOPS. The 36-chiplet Simba system is functional over a slightly narrower voltage range, from 0.52–1.1 V, achieving 0.16 pJ/op at 0.52 V and 484 MHz; at 1.1 V, the 36-chiplet system achieves a 1.8 GHz PE frequency and 128 TOPS.

3.3 Simba Baseline Tiling

To map DNN layers onto the hierarchical tile-based architecture, we first use a state-of-the-art DNN tiling strategy that uniformly partitions weights spatially, leveraging model parallelism [13, 14, 39, 51, 59, 73]. Listing 2 shows the default tiling in a loop-nest form. Each dimension of a DNN layer can be tiled temporally, spatially, or both at each level of the system hierarchy: package, chiplet, PE, and vector MAC. The loop bounds and orderings in Listing 2 are configurable in Simba so that users can flexibly map computation to the Simba system. In particular, the default dataflow uniformly partitions weights along the input channel (C) and the output channel (K) dimensions, as noted in the `parallel_for` loops. In addition, Simba can also uniformly partition along the height (P) and width (Q) dimensions of an output activation across chiplets and PEs to support flexible tiling. Section 5 highlights the limitations of this approach when mapping networks onto a large-scale, non-uniform network access architecture with an MCM-based integration.

We developed a flow that uses Caffe [38] to map a DNN inference application to the Simba system, which primarily determines an efficient tiling strategy for the dataflow that best exploits data reuse in the memory hierarchy. To facilitate evaluation of different mapping alternatives, we also developed a fast, analytical energy model for Simba that quantifies the energy cost of a particular mapping, similar to the methodology discussed in prior work [15, 25, 50]. The

compilation process starts with a *mapper* that is provided with data regarding available system resources (including the number of PEs, the number of Global PEs, and the sizes of buffers in the system) and the parameters of a given layer from the Caffe specification. The mapper determines which PE will run each portion of the loop nest and in which buffers the activations and weights are stored. As this mapping is a logical one, the mapper is followed by a *placer* which decides in which physical resource in the Simba topology the loop nests and data structures are placed. We use a random search algorithm to sample the mapping space and use the energy and performance models to select good mappings and placements. Finally, the flow generates the configuration binaries for each chiplet that implement the execution created by the mapper and placer.

4 SIMBA CHARACTERIZATION

This section details the performance characterization of Simba, focusing on achieved scalability using the uniform-tiling baseline. All evaluation results are measured using the prototype system.

4.1 Methodology

Figure 4 shows the experimental setup for measuring the performance and power of the Simba prototype system. The silicon prototype test board is attached to an x86 host through PCI-E using a Xilinx FPGA. To measure the performance of the Simba prototype system, we use software running on the RISC-V to query cycle counters built into the RISC-V microcontrollers. The runtime software designates one chiplet’s RISC-V microcontroller to be the lead RISC-V, which tracks the time from the start of execution until all chiplets complete their assigned work. Power and performance measurements begin after the weights have been loaded into each PE’s weight buffer and the inputs have been loaded into the Global PE buffers. Measurements conclude after all other partitions have signaled their completion to the lead RISC-V. Unless otherwise noted, the chiplets operate at a core voltage of 0.72 V, a PE frequency of 1.03 GHz, and GRS bandwidth of 11 Gbps. We use sense resistors on the board power supplies and a digital acquisition module to measure energy during experiment execution. Since the chiplets support independent clock frequencies for different units (PEs, Global PEs, RISC-V, and NoP routers), we can vary these frequencies to change the compute-to-bandwidth ratios for our experiments. The NoC and NoP routing tables use dimension-ordered X-Y routing for all inter-chiplet communication. Although all 36 Simba chiplets are functional, our evaluation uses 32 chiplets, as it is easier to partition computation by powers of two since the number of input channels (C) and output channels (K) are typically powers of two.

We focus our application measurements on ResNet-50 [33], a state-of-the-art, representative deep-learning network, and evaluate its layers running on the Simba system with a batch size of one, as low-latency inference is a highly critical deployment scenario for data center inferencing [32, 39, 43]. We compile and run each layer independently, except when we map multiple layers to different physical partitions of Simba and execute them in a pipelined manner. Networks are pre-trained and quantized to 8-bit using TensorRT without accuracy loss [48]. While we focus on ResNet-50 in this paper, we also present measurement results from DriveNet [10] to demonstrate Simba’s weak scaling performance, while AlexNet [41]

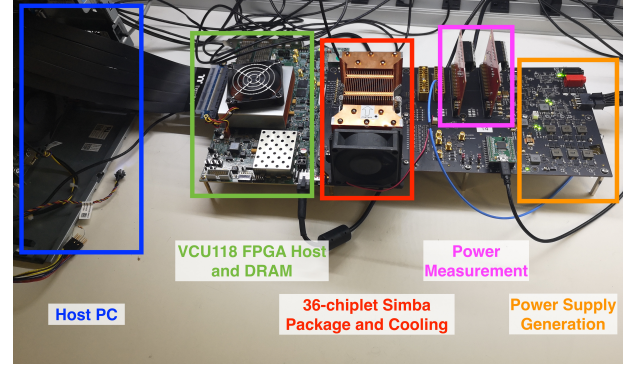


Figure 4: Bench measurement setup for the Simba prototype.

layers exhibit similar behavior. We believe that the diversity of layers in ResNet-50 provides sufficient breadth to cover a wide range of behaviors across different convolutional networks.

4.2 Overview

Figure 5 summarizes performance and energy measurements across all ResNet-50 layers. Each point represents a unique mapping for that layer, while different colors show the number of chiplets active for that mapping. Latency is normalized to a hypothetical best-achievable latency that would be realized if each of the 576 PEs of the system operated with 100% utilization and no communication or synchronization overheads. Simba provides a large number of mapping options with drastically different performance and energy profiles, highlighting the importance of strategies for efficiently mapping DNNs to hardware. The figure also demonstrates the highly variable behavior of different layers. For example, the most energy-efficient configurations of layer `res3[b-d]_branch2b` achieve almost an order of magnitude better efficiency than those of `res3a_branch2a`. The degree of data reuse highly influences the efficiency; layers with high reuse factors, e.g., the 3×3 convolution in `3[b-d]_branch2b`, tend to perform computation more efficiently than layers that require more data movement. Finally, although increasing the number of chiplets used in the system improves performance, it also leads to increased energy cost for chiplet-to-chiplet communication and synchronization. Efficiency can drop by nearly an order of magnitude for some layers, which further emphasizes the effect of data movement on overall efficiency. To better understand system-level trade-offs, the remainder of this section characterizes the sensitivity of Simba to mapping alternatives, layer parameters, bandwidth, latency, and weak scaling, and includes a comparison to modern GPUs.

4.3 Mapping Sensitivity

Figure 6 shows a performance comparison of mapping ResNet-50’s `res4[b-f]_branch2a` layer onto multiple PEs, either on-chiplet or spanning multiple chiplets. When mapped to a single chiplet, execution latency decreases linearly from one to eight PEs because of the improved compute throughput with more PEs. At the same time, its performance flattens out beyond eight PEs due to the memory bandwidth contention at the Global PE’s SRAM. However,

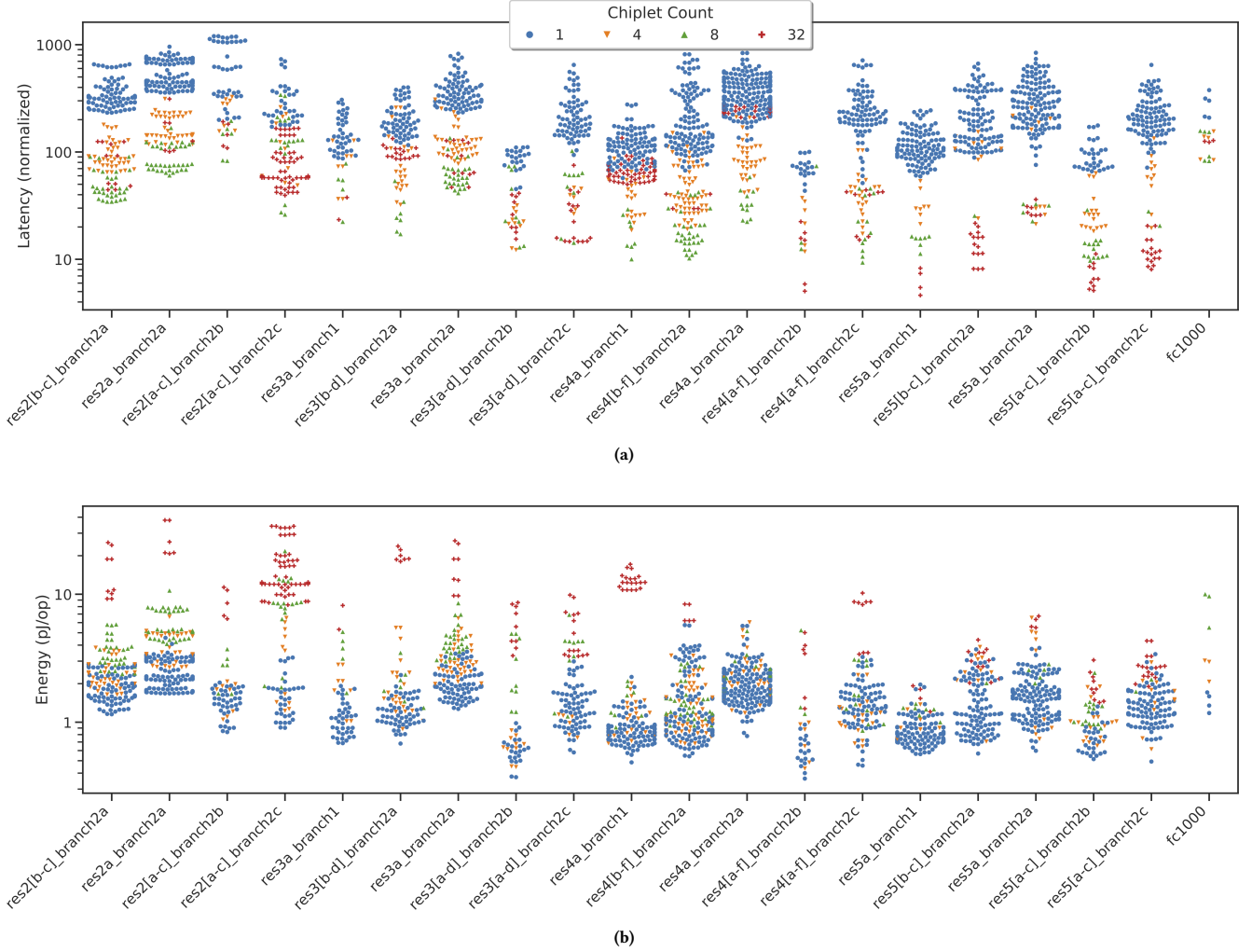


Figure 5: Measured performance and energy and running ResNet-50 on the fabricated Simba prototype. Each point is a valid workload mapping onto the system; each column cluster shows the different performance and energy achieved by different mappings of the same workload. Each symbol shape represents a different number of active chiplets used for the mapping.

when mapping across chiplets, execution time does not scale down beyond four PEs. The additional latency to communicate across multiple chiplets, including inter-chiplet communication latency and synchronization latency, ultimately leads to a $2\times$ greater execution time relative to employing a single chiplet. Good mapping strategies for MCMs must consider the different characteristics of the NoC and NoP to deliver efficient utilization of the hardware.

4.4 Layer Sensitivity

Figure 7 shows the performance scalability of running three different layers in ResNet-50 across different numbers of chiplets. While the performance of res2[a-c]_branch2b initially improves with increased chiplet count, the performance gains cease beyond

eight chiplets. As one of the early layers of the network, the number of weights in this layer is so small that it cannot fully utilize the compute throughput of Simba. The performance degrades with 32 chiplets because the inter-PE communication costs overwhelm the limited parallelism. In contrast, the performance of the res3a_branch1 layer scales to 8 chiplets, where it plateaus. This layer has more compute parallelism than res2[a-c]_branch2b, but it still does not have enough to fully overcome the overheads of inter-chiplet communication.

The res5[a-c]_branch2b layer demonstrates the best performance scaling, with improvements seen up to 32 chiplets. However, performance scaling slows down significantly past eight chiplets due to communication overheads. Of the 53 layers of ResNet-50, 12 follow the behavior of res3a_branch1, 24 follow that of res5[a-c]_branch2b, and the remaining 17 have behavior similar

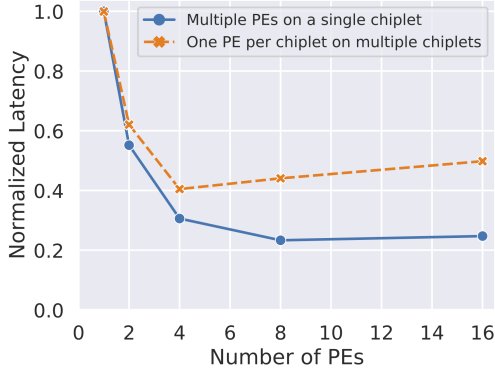


Figure 6: Performance comparison of on-chip and on-package communication and synchronization in Simba. Latency is normalized to single-PE execution latency.

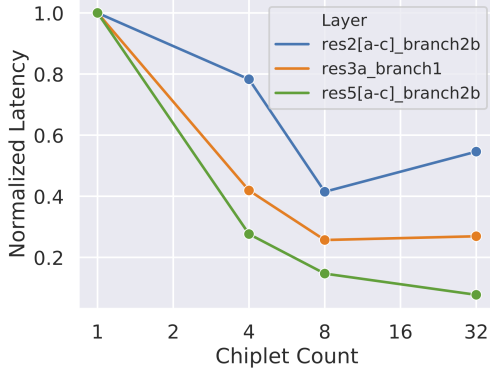


Figure 7: Simba scalability across different layers from ResNet-50. Latency is normalized to the latency of the best-performing tiling with one chiplet.

to res2[a-c]_branch2b. These measurements demonstrate that the amount of compute parallelism that an MCM can leverage varies from layer to layer, and that the cost of communication can hinder the ability to exploit that parallelism, even on a single chiplet.

4.5 NoP Bandwidth Sensitivity

To examine the bandwidth sensitivity of different layers, we adjust the bandwidth of the NoP relative to the intra-chiplet compute performance of the system. This adjustment is made by reducing the frequencies of the PE, Global PE, and RISC-V partitions below nominal while maintaining a constant NoP frequency. By measuring results in terms of the number of PE cycles required for computation, this frequency reduction effectively corresponds to an increase in NoP bandwidth. Figure 8 shows how execution time is affected by NoP bandwidth for two representative ResNet layers when mapped to 32 chiplets. For res3[a-d]_branch2b, the increased bandwidth between chiplets results in only a 5% decrease in execution time, indicating that this layer is not bound by NoP bandwidth or inter-chiplet communication latency. However, for

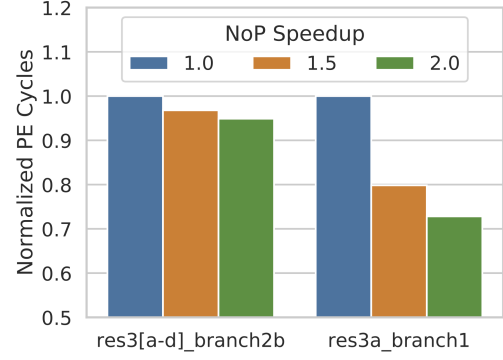


Figure 8: Simba scalability with different chiplet-to-chiplet communication bandwidths.

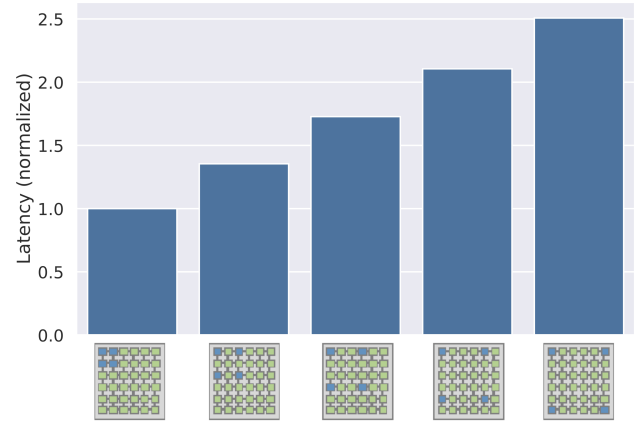


Figure 9: Simba scalability with different chiplet-to-chiplet communication latencies running res4a_branch1 with four chiplets (using the same tiling). Different bars represent different selections of the active four chiplets, as shown under the X-axis; the active chiplets are highlighted in blue.

res3a_branch1, the increased bandwidth decreases execution time by 27%, indicating that this layer is bottlenecked by communication between chiplets. Because an MCM-based system intrinsically has a NUNA architecture between intra-chiplet and inter-chiplet PEs, mapping policies must consider the different latency and bandwidth parameters to deliver good performance and efficiency.

4.6 NoP Latency Sensitivity

In addition to lower bandwidth, the NoP has higher latency than the NoC due to inter-chiplet signaling overheads. To isolate the effect of NoP latency, we ran experiments mapping layers to four chiplets, but adjusted the locations of the selected chiplets in the package to modulate latency. Figure 9 shows the effect of increasing the longest inter-chiplet latency from 2 hops to 12 hops for res4a_branch1. The figure shows the execution time normalized to a configuration of adjacent chiplets, with the chiplet selection shown under each bar. With active chiplets further apart, the overall execution time

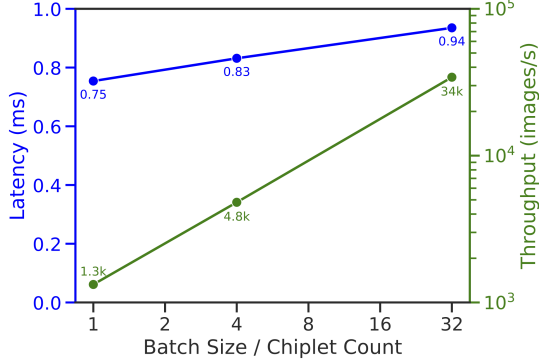


Figure 10: Characterization of weak scaling on Simba running DriveNet.

increases by up to $2.5\times$ compared to execution on adjacent chiplets. Communication latency is typically less pronounced for small-scale systems but plays a significant role in achieving good performance and energy efficiency for a large-scale, MCM-based system like Simba.

4.7 Weak Scaling Sensitivity

Figure 10 shows the weak scaling trend of Simba running DriveNet, an end-to-end DNN for self-driving cars [10]. As DriveNet is a significantly smaller network than ResNet-50, a one-chiplet mapping is the most energy-optimal configuration for batch size 1. We use DriveNet to demonstrate the weak scaling trend of Simba. Instead of distributing the same amount of computation to multiple chiplets, as in the case of ResNet-50, we fix the amount of work per chiplet but increase the total amount of computation by increasing the batch size. Figure 10 shows that by increasing the number of active chiplets from one to 32, Simba achieves a $26\times$ throughput improvement (compared to $32\times$ for perfect weak scaling), while incurring a 24% latency increase due to synchronization cost across multiple chiplets.

4.8 Comparisons with GPUs

Figure 11 compares Simba to NVIDIA’s V100 and T4 GPUs. We run ResNet-50 with different batch sizes and compare to the GPU results published in [49]. Due to Simba’s limited on-package storage capacity for input activations, we only run Simba at batch size one and two. Unlike GPUs, Simba is designed for low-latency inference with a small batch size, which motivates the use of distributed and persistent weight storage to reduce data movement. The Simba package, including the MCM interface, has substantially smaller total silicon area (216 mm^2) than (525 mm^2) or V100 (815 mm^2), due to differences in math precision, on-chip storage, DRAM interface, and types of computation supported in these architectures.

Figure 11a shows the throughput of Simba, V100, and T4 running ResNet-50. Simba delivers $1.8\times$ and $1.9\times$ better throughput at batch size one compared to V100 and T4, respectively. Figure 11b illustrates the corresponding energy efficiency improvement of Simba compared to V100 ($5.4\times$) and T4 ($2.9\times$). When running ResNet-50

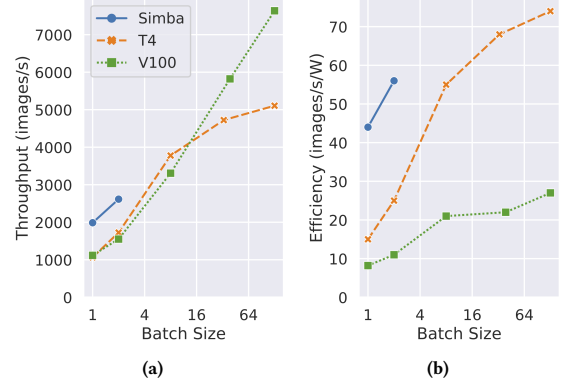


Figure 11: Throughput and efficiency of Simba, V100, and T4 running ResNet-50 with different batch sizes.

with a larger batch size, instead of exploiting the batch-level parallelism like GPUs, Simba would run each batch sequentially. As a result, we expect the throughput of Simba is close to that of running with a batch size of one.

5 SIMBA NON-UNIFORM TILING

This section presents three novel DNN workload tiling techniques that target the non-uniform latency and bandwidth presented by large-scale MCM-based systems. In all three cases, we stress the importance of communication-latency-aware tiling when mapping DNN workloads to large-scale, hierarchical systems.

5.1 Non-Uniform Work Partitioning

Efficient use of parallel systems requires proper load balancing among the components in the system. Failure to properly balance the load on the system leads to higher latency and energy caused by resources waiting for the slowest unit to complete, i.e., increased tail latency. The total execution time can be broken down into two major components: communication latency and compute latency. State-of-the-art DNN tiling strategies typically assign the same amount of the work to each of the available resources [71]. However, this approach breaks down for large-scale systems, especially when the PEs are spatially distributed with different communication latencies between them.

To address this limitation, we propose a non-uniform work partitioning strategy that considers communication latencies. Instead of uniformly assigning the same amount of work to each PE, we non-uniformly partition the work across the PEs. PEs closer to the data producers will perform more work to maximize physical data locality, while PEs that are further away will do less work to decrease the tail latency effects. Figure 12 illustrates an example of non-uniform work partitioning using a 4-chiplet system. In this example, we assume the input activation (IA) is physically stored in the Global PEs of Chiplet0 and Chiplet2, while the weights (W) and the work is partitioned across all four chiplets. During execution, the Global PE of Chiplet0 will multicast a slice of IA

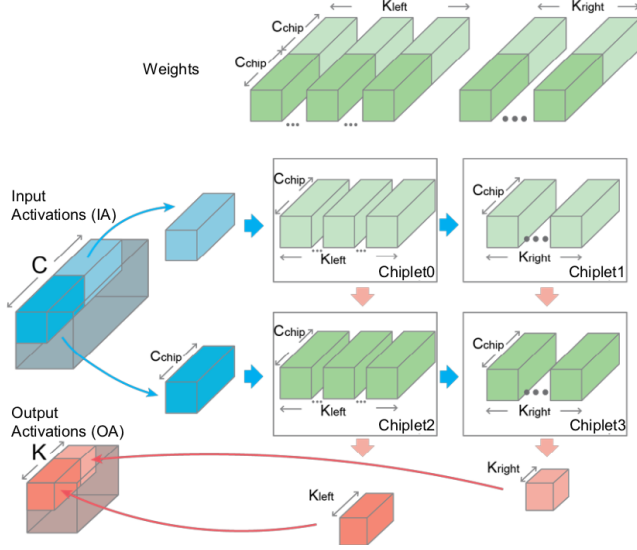


Figure 12: Illustration of communication-aware, non-uniform work partitioning. The top green tensors represent weights (W), the left blue tensors represent input activations (IA), and the bottom red tensors represent the output activation (OA). In this example, IA is stored in Chiplet0 and Chiplet2.

to PEs in both Chiplet0 and Chiplet1. Because the communication latencies from the Chiplet0 Global PE to the PEs in Chiplet0 and Chiplet1 are different, Chiplet1 will fall behind. To prevent the longer communication to Chiplet1 from increasing the tail latency of the execution, we can adjust the amount of computation that each chiplet is assigned in a manner inversely proportional to its communication distance from the source. In the example shown in Figure 12, Chiplet0 and Chiplet2 are provided with larger chunks of work (K_{left}) while Chiplet1 and Chiplet3 get the smaller chunks (K_{right}). This work schedule evens out the completion time across the chiplets, thereby improving overall system performance. For simplicity, this example only shows non-uniform partitioning with respect to input activations. A similar technique can be used to mitigate the communication latency for output activations to the destination chiplets by using non-uniform partitioning along the C dimension.

The variation in communication latency is quite pronounced in large-scale systems such as Simba, with hundreds of spatially distributed PEs. To dynamically adjust the amount of work that each PE performs, we use the performance counters within each PE to collect accurate latency and utilization information during the initial execution of a layer. We then adjust work distribution for subsequent executions of each layer based on the latency variation across PEs.

Figure 13 illustrates the measured performance improvement using non-uniform work partitioning. For each of the layers, we pick the highest-performance uniform tiling from Figure 7 as the baseline. We then measure the execution time of different chiplets and identify layers with a large tail latency. For these layers, we

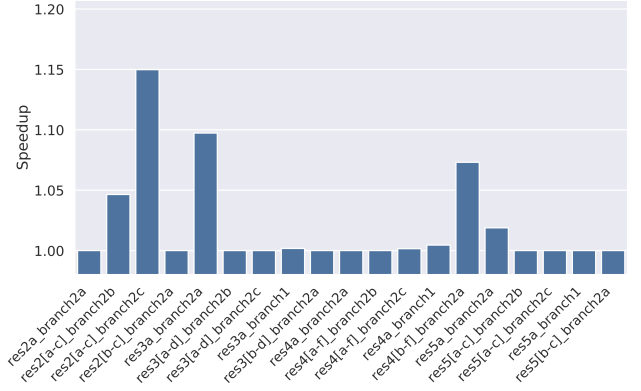


Figure 13: Non-uniform work partition for ResNet-50 with speedup normalized to the best-performing tiling.

use non-uniform work partitioning to shift the computation from the tail PEs to the PEs that are closer to the data. Depending on layer dimensions, we achieve up to 15% performance improvement compared to the best uniform tiling for a given layer. We notice that the achievable performance improvement is highly sensitive to the compute-and-communication ratio in a given mapping. For example, when either compute or communication is significantly dominating the overall execution latency, as in the case of res5a_branch1, incrementally modulating the amount of work each PE performs provides little performance improvement. However, when compute and communication latencies are more comparable, which is typically desired to achieve good mapping, the performance improvement is more pronounced, as in the case of res2[b-c]_branch2c.

5.2 Communication-Aware Data Placement

The communication latency in a parallel system can have a large effect on the overall system performance, as observed in multi-core and multi-GPU characterizations [5, 8, 9, 16, 31, 35, 47, 66–68]. Due to the limited scale of today’s deep-learning accelerators, most have one unified global buffer that supplies data to all of the PEs [1, 13, 15, 59, 75]. However, in large-scale MCM system where on-chip buffers are spatially distributed among the chiplets, communication latency becomes highly sensitive to the physical location of data. Figure 14 illustrates how data placement affects communication distances and latencies. For example, if the Src chiplet in Figure 14(a) broadcasts data to the other chiplets, the arrival time of the data will vary greatly depending the distance of the receiving chiplets from Src. Depending on the amount of computation each chiplet performs, such variations in communication distance could significantly limit the achievable speedup in a distributed, tile-based system like Simba, motivating the need for data placement optimization.

While optimal data placement is an NP-hard problem, we use a practical greedy algorithm to iteratively determine where input and output activation data should be placed in the Simba system. The algorithm starts by performing placement of the input activation data blocks. Once the input activations are placed, the same greedy algorithm is executed for tiles of output activations. Since a

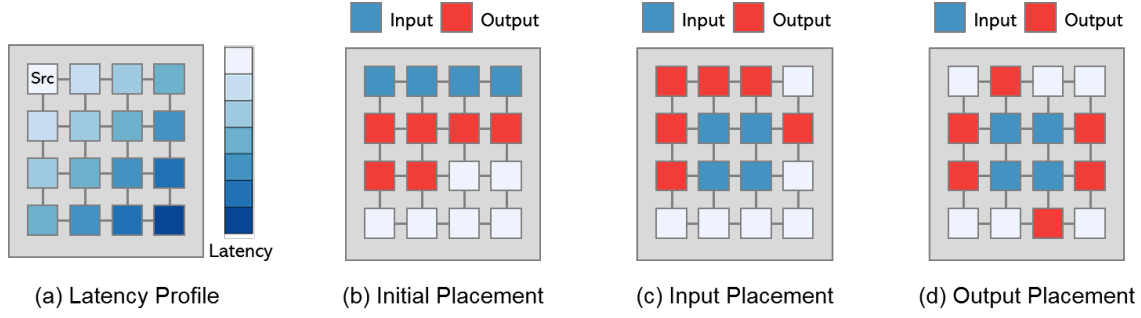


Figure 14: Data placement on the Simba system. (a) Assessment of the relative latency to different chiplets that receive data from Src. (b) Default input activation (IA) and output activation (OA) placement where data is sequentially placed from the Global PE of the first chiplet. (c) An improved IA placement at the center of the package so that data can be multicast to all chiplets. (d) OA placement with even distribution along the periphery of the package to minimize OA communication latency.

previous stage of the mapping process has already determined the data tiling, this stage need only focus on data placement and not re-tiling. Figure 14(b) shows a naive data placement with a sequential allocation of input activations to the chiplets on the top row and output activations in the next six chiplets. Figure 14(c) shows a better assignment of IAs to chiplets, selecting the four in the

Algorithm 1: Iterative data placement algorithm.

Result: Determine placement of the input activation data

Select an input activation block;

Precompute communication cost between different source-destination pairs

while not the end of input activation do

for each possible chiplet placement do

 Calculate communication cost using pre-computed look-up table;

 Select a chiplet that minimizes the communication cost;

if that chiplet's RAM is full then

 Select the next best source chiplet;

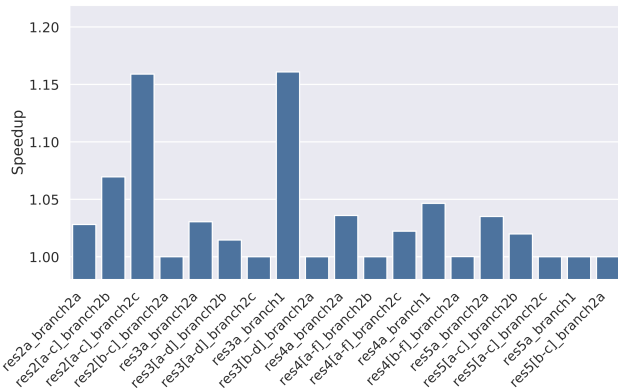


Figure 15: Data placement for ResNet-50 layers with speedup normalized to the best performing tiling.

middle that minimize aggregate multi-cast hop-count to all chiplets. Finally, Figure 14(d) shows a placement of output activations on chiplets in regions where OA accumulation can occur.

Figure 15 shows the performance improvement of ResNet-50 layers with optimized data placement. Although all of the layers use 32 chiplets, many of them have different communication patterns. For example, layers like res2[a-c]_branch2c communicate frequently within a group of eight chiplets. In this case, it is better to group those chiplets together to minimize communication cost. In contrast, layer res4a_branch1 must broadcast from a single chiplet to all 32 chiplets. In this case, instead of placing the source chiplet sequentially at the upper left corner, placing it at the center of the package leads to a 5% performance improvement. Data placement optimization results in up to 15% improved performance compared to the best achieved baseline.

5.3 Cross-Layer Pipelining

One key challenge of mapping DNN layers to large-scale systems is achieving high utilization when the layer computation has a limited amount of parallelism [14, 39, 71]. To address this challenge, recent DNN accelerators support pipelined execution to improve overall system utilization [26, 71]. ScaleDeep supports column-wise pipelining in the PE array, where different columns can be assigned to different pipelined layers [71]. This low-overhead implementation still results in low utilization when layers cannot be easily mapped across columns. Tangram supports flexible partitioning across the PE array for different layer shapes but does not consider the non-uniformity of communication latency and bandwidth [26].

Figure 16 illustrates how a residual block of ResNet-50 can be pipelined across the Simba package. Because the Simba hierarchical interconnect supports flexible communication patterns, we can assign different-sized clusters of chiplets to different layers. In the example shown, res2a_branch2b uses four chiplets, while res2a_branch2a uses only two chiplets. Figure 17 shows the achieved throughput improvement from pipelining three residual blocks. Within each residual block, instead of executing each layer sequentially on the entire package, we partition the package into three or four clusters, assign different layers to each cluster, and execute the layers in a pipelined fashion. Pipelined execution

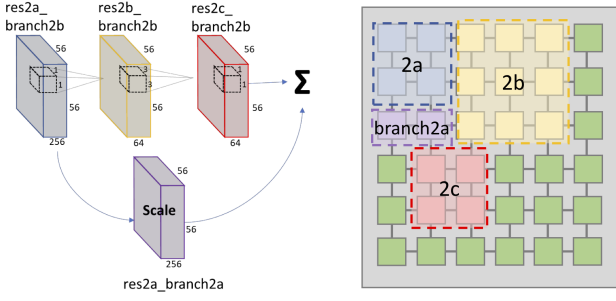


Figure 16: Pipelining a residual block of ResNet-50 in the Simba system.

on Simba improves overall throughput by up to $2.3\times$ compared to the sequential execution baseline. With pipelining, the overall throughput is limited by the longest pipelining stage. For example, `res3_x` achieves the most significant speedup due to the relatively balanced sizes of input activations and weights, while the earlier `res2_x` layers have much larger input activations than weights. Later `res4_x` layers are more dominated by weight size, leading to less throughput improvement.

6 RELATED WORK

DNN inference applications typically run on highly programmable but inefficient CPUs, programmable GPUs with ISA extensions for accelerating tensor operations, or fixed-function DNN inference accelerators. In this work, we present an MCM-based fixed-function DNN inference accelerator prototype system that is capable of running highly complex DNN models at high throughput and low latency. Our aim for Simba is to explore scaling challenges and opportunities without incurring higher inference latency.

Many previous papers have developed hardware accelerator architectures that focus on fast and efficient execution of DL inference [2, 13, 15, 30, 51, 52, 57, 59, 61]. In some cases, these accelerators target small-scale networks and do not consider the challenges associated with scaling to larger networks. In other cases, a proposed accelerator requires structural changes to scale to large

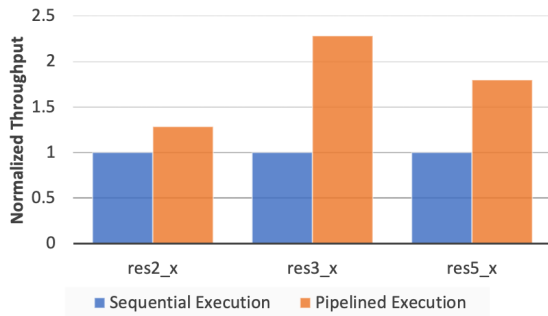


Figure 17: Throughput improvement from pipelined execution on three residual blocks of ResNet-50.

high-performance chips. For example, DianNao was originally proposed in the context of small DNN layers [13]. Follow-on work with DaDianNao aimed at larger networks by proposing to use a multi-chip network and eDRAM for weights and activations [14]. In contrast, Simba is designed from the ground up for large-scale inference, specifically employing package-level integration.

The TPU [39] is a data-center inference accelerator which lacks the mapping flexibility targeted in Simba. The TPU restricts communication to column-wise or row-wise multicast in its systolic computation core, while the communication flexibility in Simba provides the opportunity to perform data-locality-based mapping optimizations. MAERI [42] is a DNN accelerator in which the PEs are controlled by switches at run time; this work does not explore large scale inference with MCMs or chiplets as we do with Simba. MCMs have been explored in CPU design [37, 40, 45, 62, 74] and GPU design [3, 18] to circumvent the fabrication and design costs associated with producing large monolithic chips. Our contribution in Simba is the design, implementation, and evaluation of the first MCM-based DNN accelerator.

Previous work explored efficient data movement for multi-chip systems on general-purpose workloads [66]. Our work focuses on efficient data movement on MCM-based systems for DNN inference. Tangram [26] provides efficient mappings for tile-based accelerators using shared buffers, loop ordering, and application pipelining. Future work could apply Tangram’s mapping techniques to Simba architectures.

7 CONCLUSIONS

This work presents Simba, a scalable MCM-based deep-learning inference accelerator architecture. Simba is a heterogeneous tile-based architecture with a hierarchical interconnect. We developed a silicon prototype system consisting of 36 chiplets that achieves up to 128 TOPS at high energy efficiency. We used the prototype to characterize the overheads of the non-uniform network of an MCM-based architecture, observing that load imbalance and communication latencies contribute to noticeable tail-latency effects. We then showed how considering the non-uniform nature of system can help improve performance through techniques such as non-uniform work partitioning, communication-aware data placement, and cross-layer pipelining. Applying these optimizations results in performance increases of up to 16% compared to naive mappings.

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